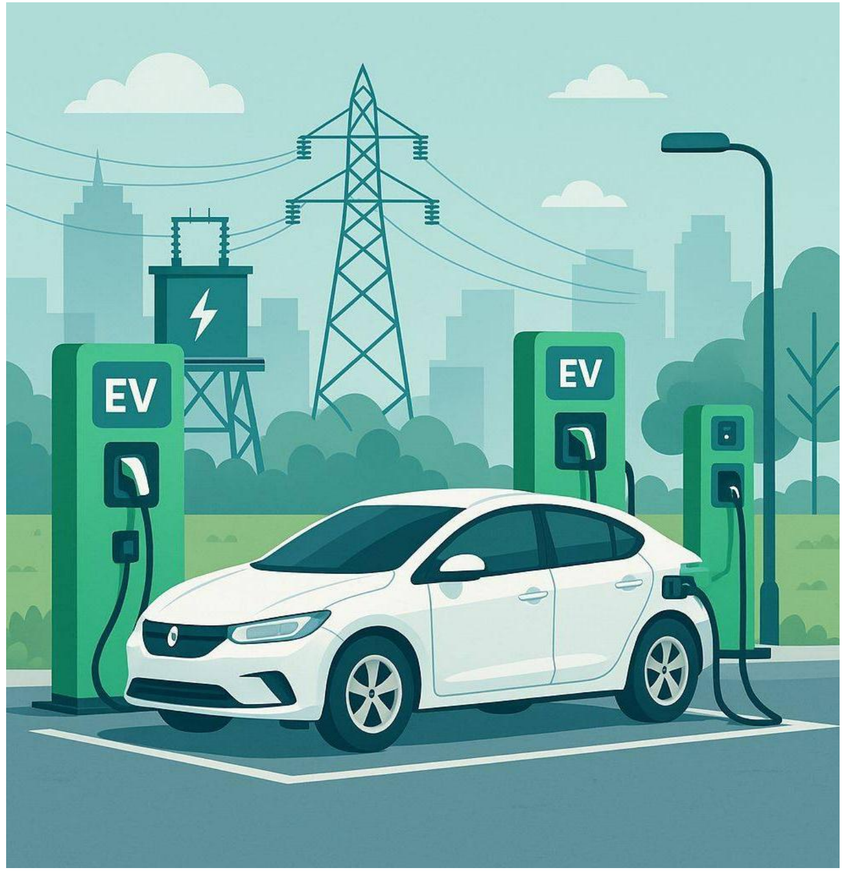
AtliQ motors

A Data-Driven Blueprint for AtliQ Motors' India Entry .
Unlocking the world's most dynamic EV market through strategic disruption.

DETECT. ANALYZE. AUTOMATE.

Problem Statement

- AtliQ Motors is a leading automotive company based in the USA, specializing in **electric and hybrid vehicles**
- Over the past 5 years, the company has achieved a **25% market share in the North American EV segment**, establishing itself as a key industry player
- As part of its global expansion strategy, AtliQ Motors plans to **enter the Indian market**, where its current market share is **below 2%**
- To support this expansion, **Bruce Haryali**, Head of AtliQ Motors India, has initiated a **comprehensive analysis of the Indian EV market**



Objective & Scope of Analysis

Objective

• To conduct a **comprehensive market study** of the EV/Hybrid industry in India and gain insights into current market dynamics.

Scope of Analysis

• EV sales trends (2022–2024)
• Penetration rates across states
• Market trends & growth patterns
• Key business insights for expansion

Data collection

Category	Source	Data Collected	Purpose in Project
EV Sales Data	Vahan Sewa	State-wise & manufacturer-wise EV sales	Trend analysis, top & bottom makers
EV Industry Data	Society of Indian Automobile Manufacturers	Industry sales reports, market share	Market comparison & validation
Market Insights	India Brand Equity Foundation	EV growth trends, sector insights	Understanding market growth
Policy & Subsidies	FAME II Scheme	Central EV incentives	Impact of subsidies on adoption
State Policies	Delhi, Maharashtra, Tamil Nadu, Gujarat	State-level EV subsidies	State-wise adoption analysis
Charging Infrastructure	Ministry of Heavy Industries	Charging station data	Infrastructure impact analysis
Infra Data Platform	e-AMRIT	Public charging station details	Supporting infra insights
Research Reports	NITI Aayog, International Energy Agency	EV projections, CAGR, penetration	Future trend analysis

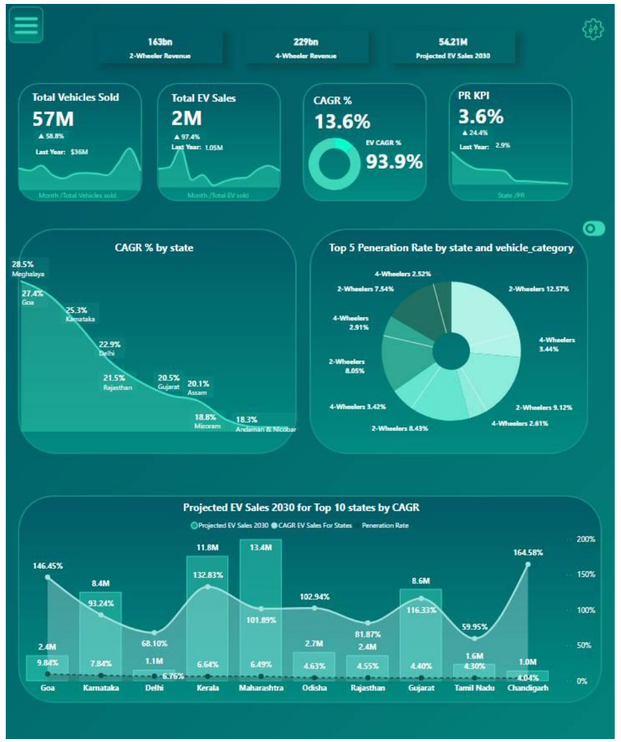
VAHAN
NR e-Services

SIAM
Society of India Automobile Manufactures
Building the Nation, Responsibly

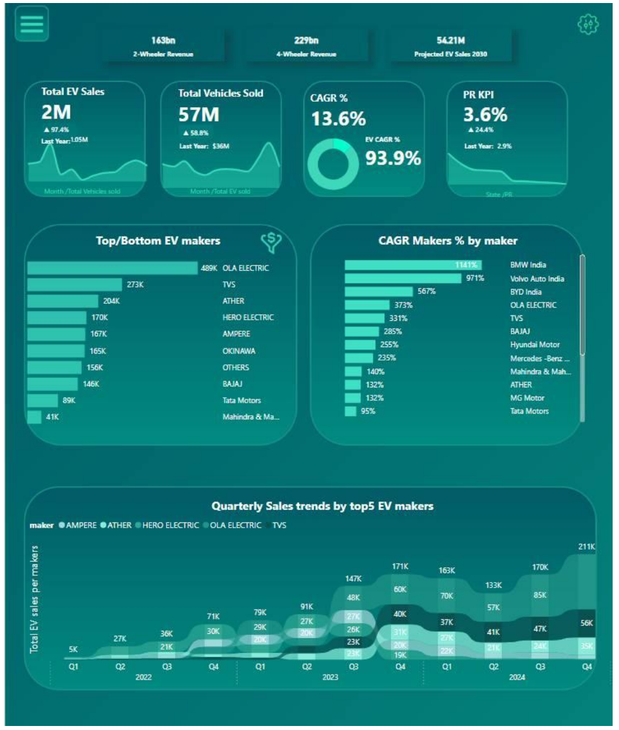
IBEF
INDIA BRAND EQUITY FOUNDATION

Dashboard overview

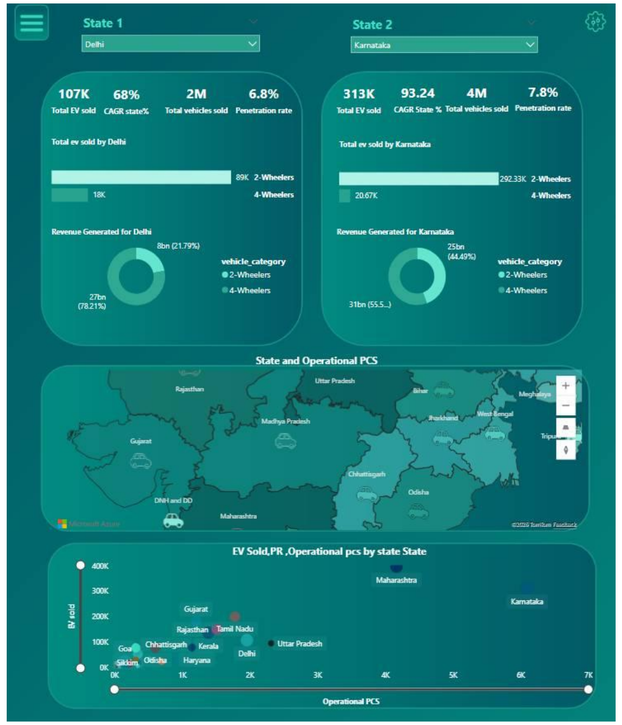
State analysis

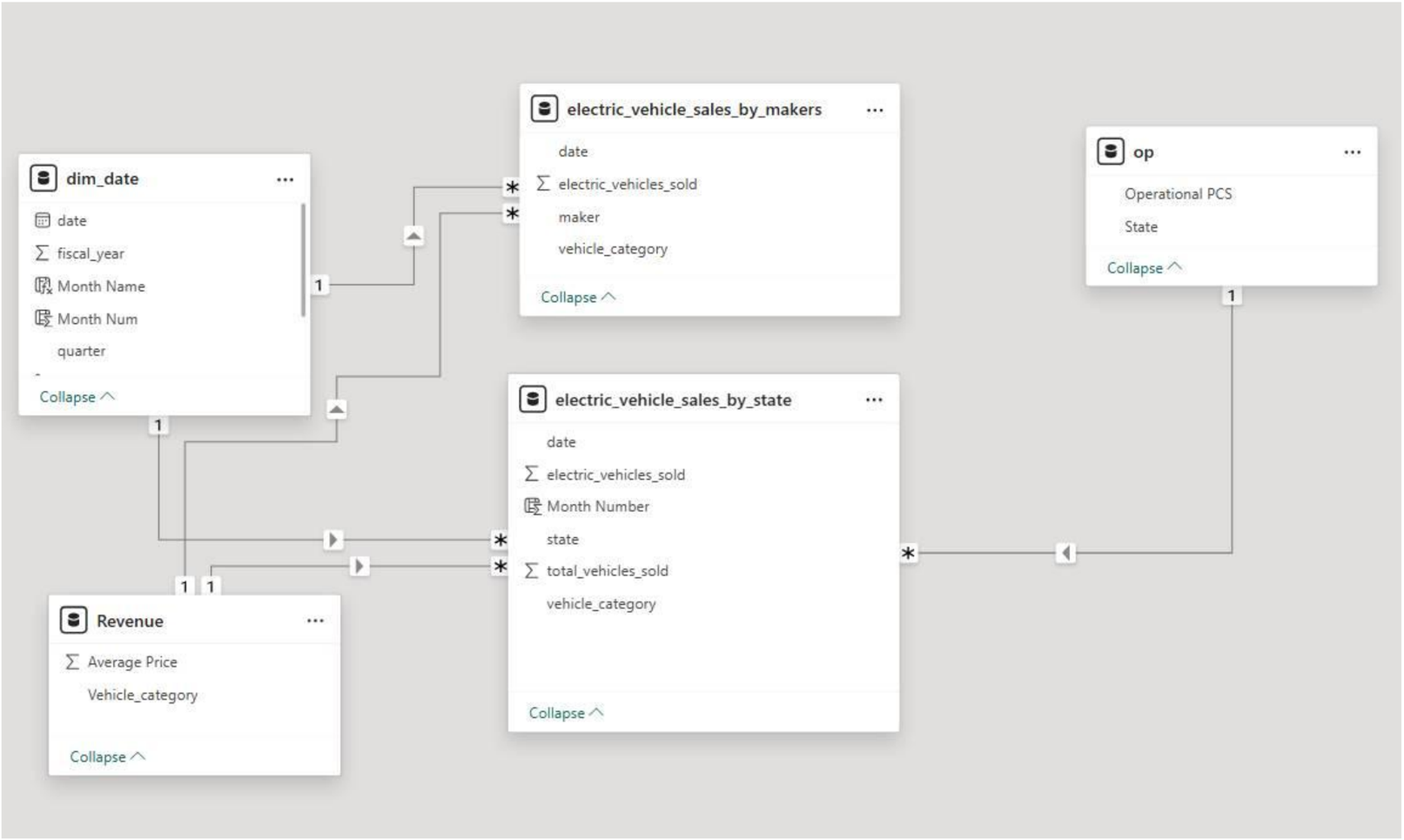


Maker analysis



State comparision

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Data Model

Primary Questions

Q1: Top & Bottom 3 manufacturers (2W sales) – FY 2023 & 2024

Q2: Top 5 states by EV penetration (2W & 4W) – FY 2024

Q3: States with negative EV penetration growth (2022–2024)

Q4: Quarterly sales trends – Top 5 EV makers (4W) (2022–2024)

Q5: EV sales & penetration comparison – Delhi vs Karnataka (2024)

Q6: CAGR analysis – Top 5 makers (4W) (2022–2024)

Q7: Top 10 states by CAGR in total vehicle sales (2022–2024)

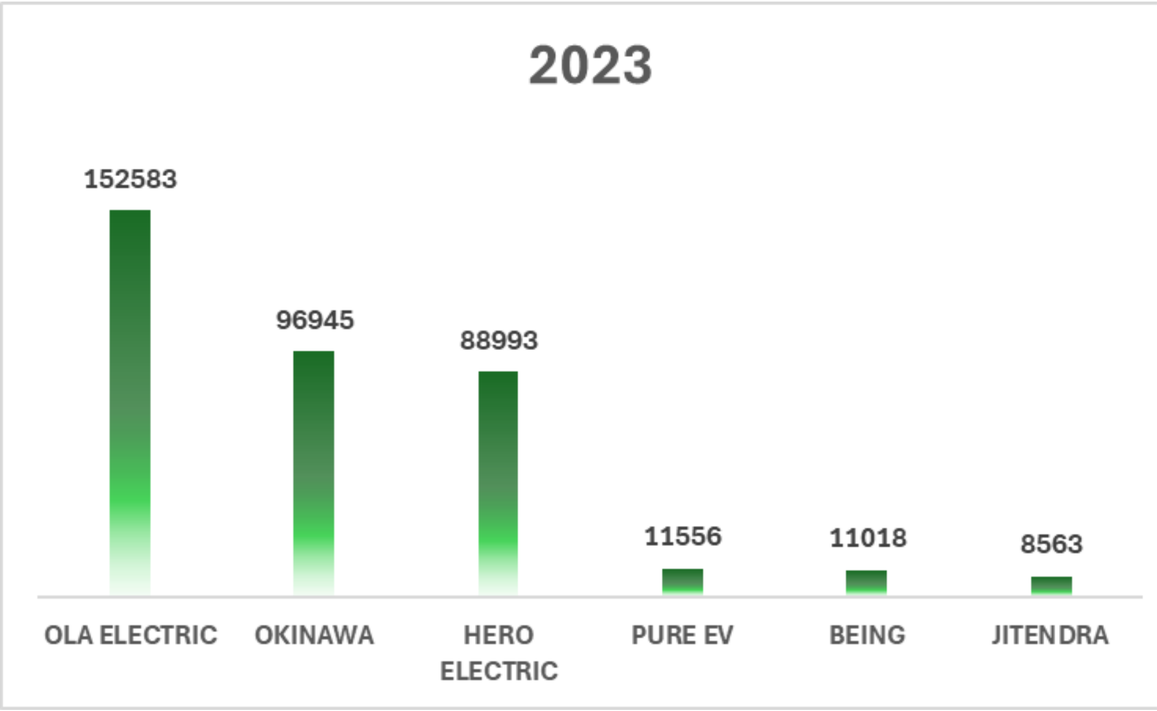
Q8: Peak & low EV sales months (2022–2024)

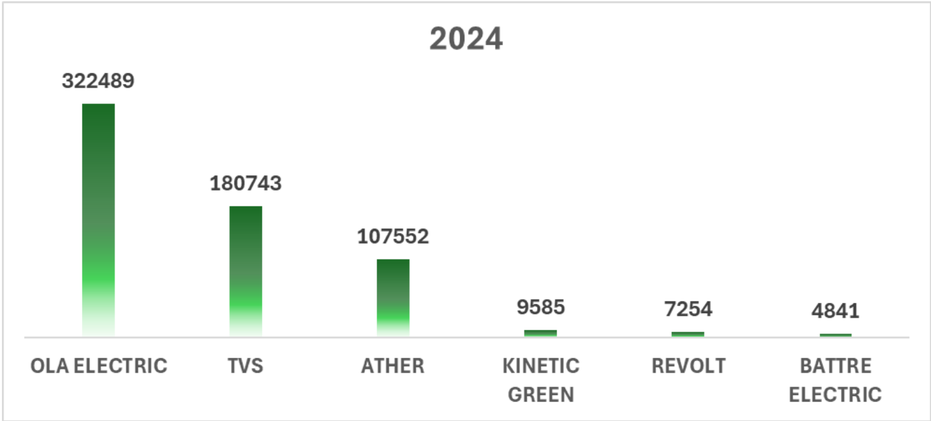
Q9: EV sales projection for 2030 (Top 10 states by penetration)

Q10: Revenue growth analysis (2W & 4W EVs)

Qn 1 : List the top 3 and bottom 3 makers for the fiscal years 2023 and 2024 in terms of the number of 2-wheelers sold.

- OLA Electric dominates both years with strong growth
- TVS and Ather entered top 3 in 2024
- Hero Electric dropped out of top 3 in 2024
- Bottom players show very low and declining sales
- Market is becoming highly competitive and concentrated





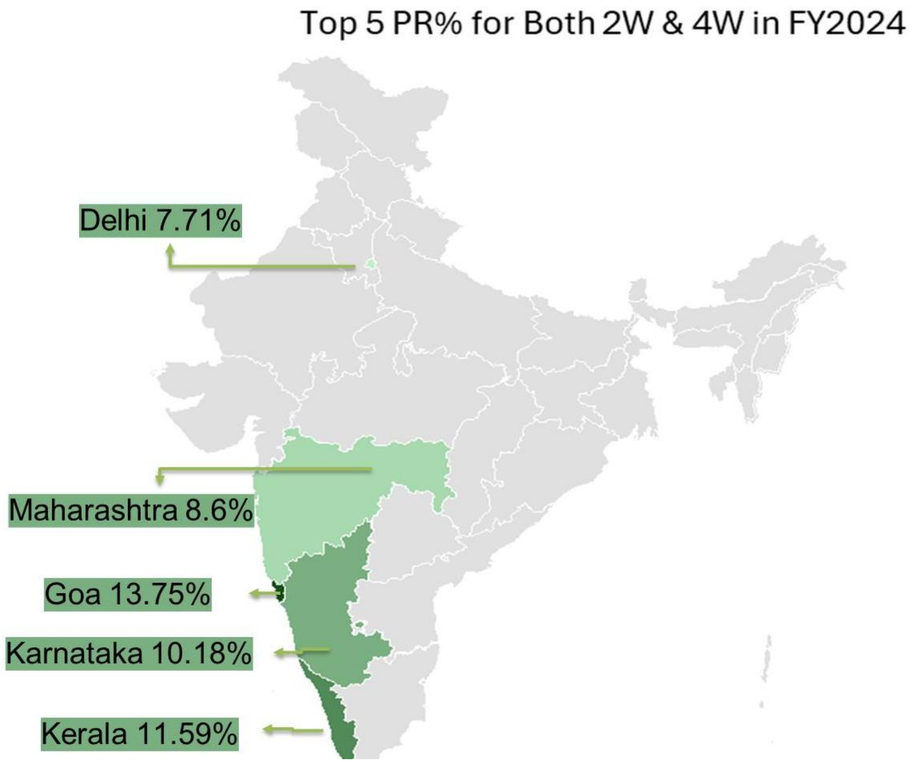
Qn 2: Identify the top 5 states with the highest penetration rate in 2-wheeler and 4-wheeler EV sales in FY 2024.

Goa leads with the highest EV penetration (13.75%)

Southern states like Karnataka & Kerala show strong adoption

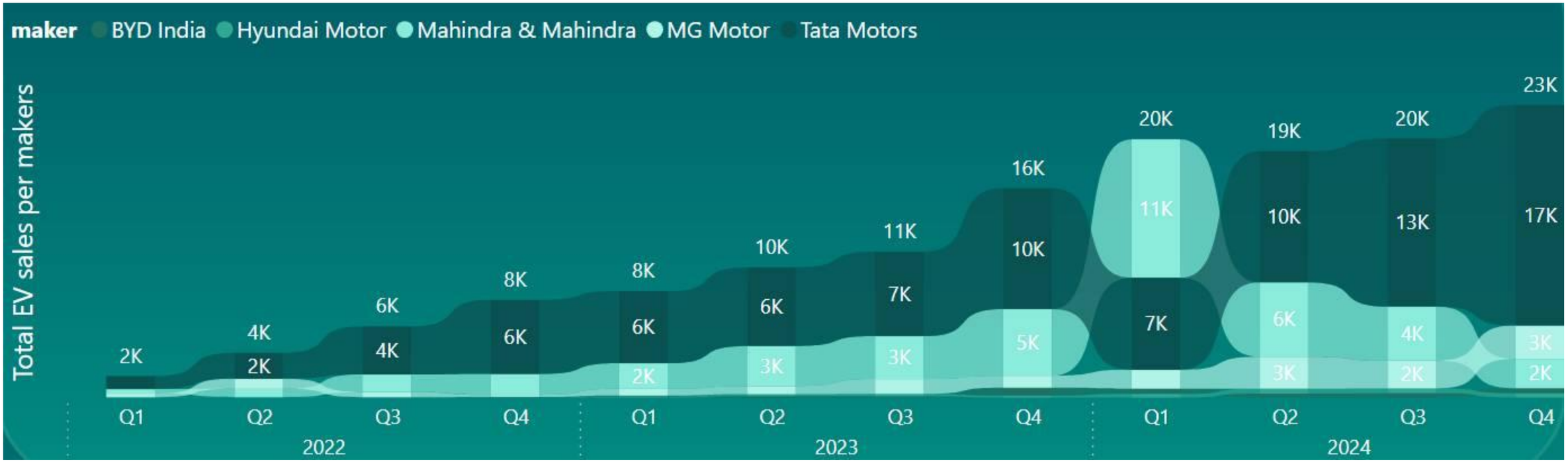
Large markets like Maharashtra & Delhi have lower penetration %

Indicates untapped growth potential in metro markets



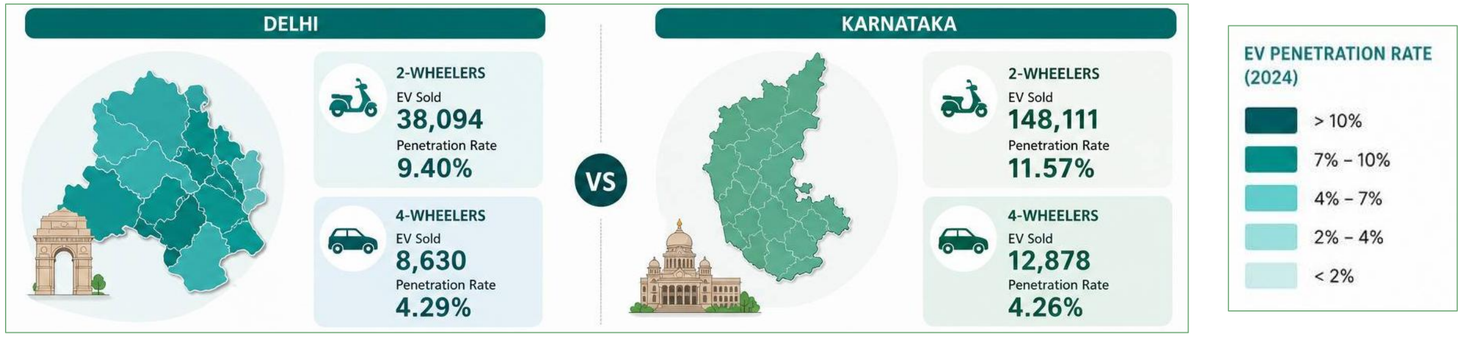
Q4: Quarterly sales trends – Top 5 EV makers (4W) (2022–2024)

- Strong upward growth across all quarters from 2022 to 2024, with 2024 marking a rapid acceleration phase in EV adoption
- Tata Motors dominates the market, showing significantly higher sales volumes and consistent leadership throughout all quarters
- Clear seasonality observed, with Q4 peaking each year, supported by festive demand and year-end purchase trends



Q5: EV sales & penetration comparison – Delhi vs Karnataka (2024)

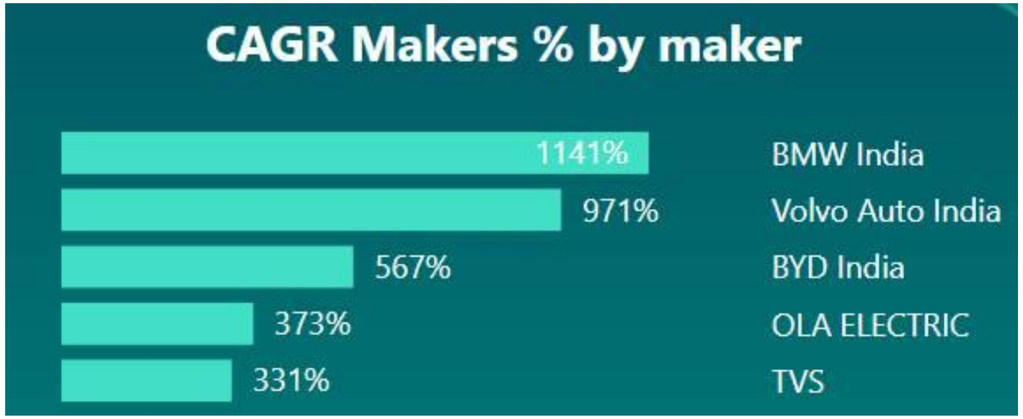
- Karnataka leads in 2-wheeler EV adoption with a penetration rate of 11.57%, compared to 9.40% in Delhi
- EV sales volume is significantly higher in Karnataka across both segments, indicating a larger and more mature EV market
- 4-wheeler EV penetration is nearly equal in both states (~4.3%), showing similar adoption levels in the premium segment
- Delhi shows strong adoption despite lower total volumes, indicating high potential in a dense urban market
- Both states demonstrate strong growth potential, with increasing adoption driven by infrastructure, awareness, and policy support



State	Vehicle Category	Total EV Sold	Total Vehicles Sold	Penetration Rate
Delhi	2-Wheelers	38,094	405,218	9.40%
	4-Wheelers	8,630	201,130	4.29%
Karnataka	2-Wheelers	148,111	1,279,767	11.57%
	4-Wheelers	12,878	302,221	4.26%

Q6: CAGR analysis – Top 5 makers (4W) (2022–2024)

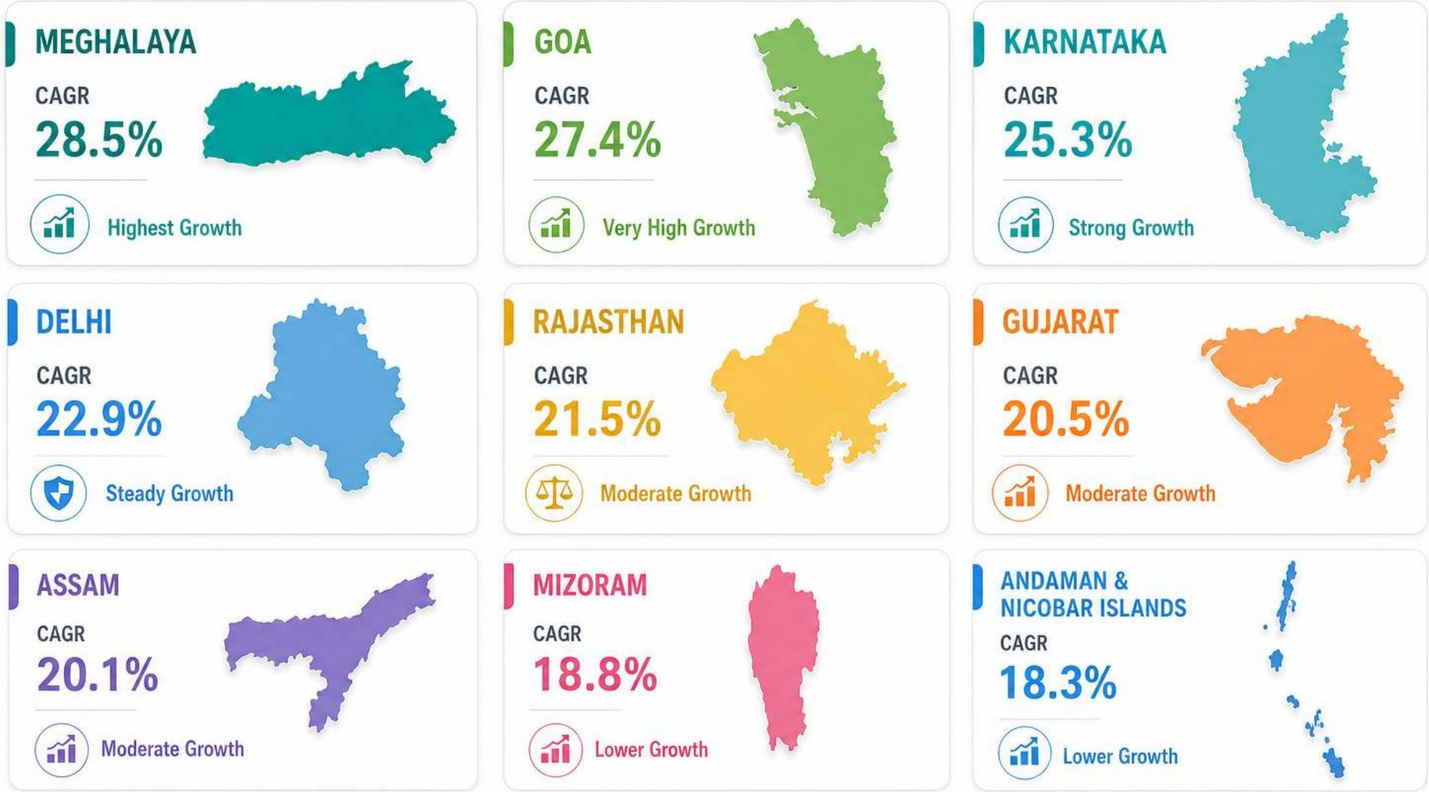
- BMW India leads with an exceptional CAGR of 1141%, indicating explosive growth in the premium EV segment
- Volvo Auto India follows closely (971%), reinforcing strong momentum in luxury EV adoption
- BYD India shows robust growth (567%), highlighting increasing competition from global EV players
- Mass-market players like OLA ELECTRIC and TVS Motor Company show steady but lower CAGR, indicating more stable, volume-driven growth
- Overall trend shows rapid expansion across all players, with premium segment growing faster than mass market



CAGR by State (2022–2024)

Q7: Top 10 states by CAGR in total vehicle sales (2022–2024)

- Meghalaya leads with the highest CAGR (28.5%), followed by Goa and Karnataka, indicating strong growth in emerging and coastal markets
- Mid-tier growth observed in Delhi, Rajasthan, and Gujarat, reflecting stable expansion in established markets
- North-East states like Assam and Mizoram show consistent growth, highlighting rising EV adoption beyond metro regions
- Lower CAGR in Andaman and Nicobar Islands and Mizoram, indicating slower but steady adoption phases
- Overall trend shows EV growth expanding from metro to emerging regions, signaling market diversification and future scalability

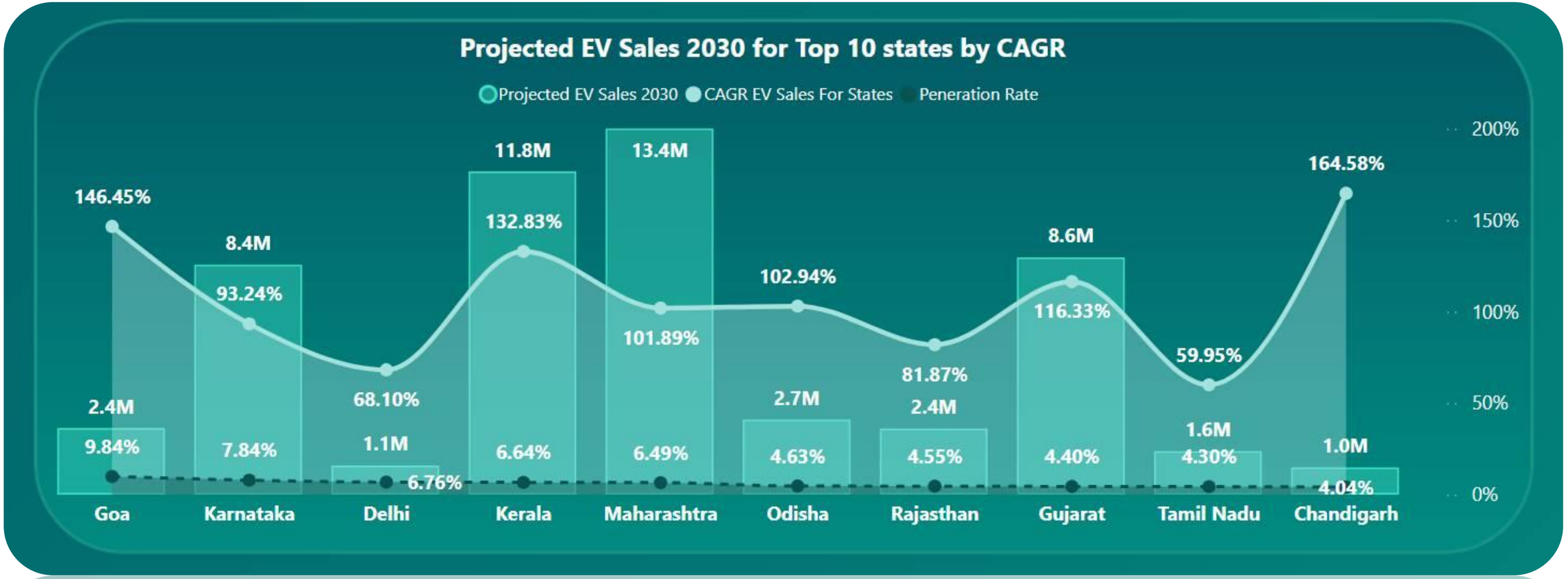


Q8: Peak & low EV sales months (2022–2024)

- EV sales increase significantly from 2022 → 2024, showing rapid market expansion and rising adoption
  Sales generally start lower in early months and increase toward year-end, indicating a clear growth pattern
  Sales peak during October–March, driven by festive demand, offers, and financial year-end purchases
  March records the highest sales across all years *(peak: Mar 2024 – 1,38,343)*
  Especially May shows the lowest sales *(lowest: May 2022 – 1,499)*

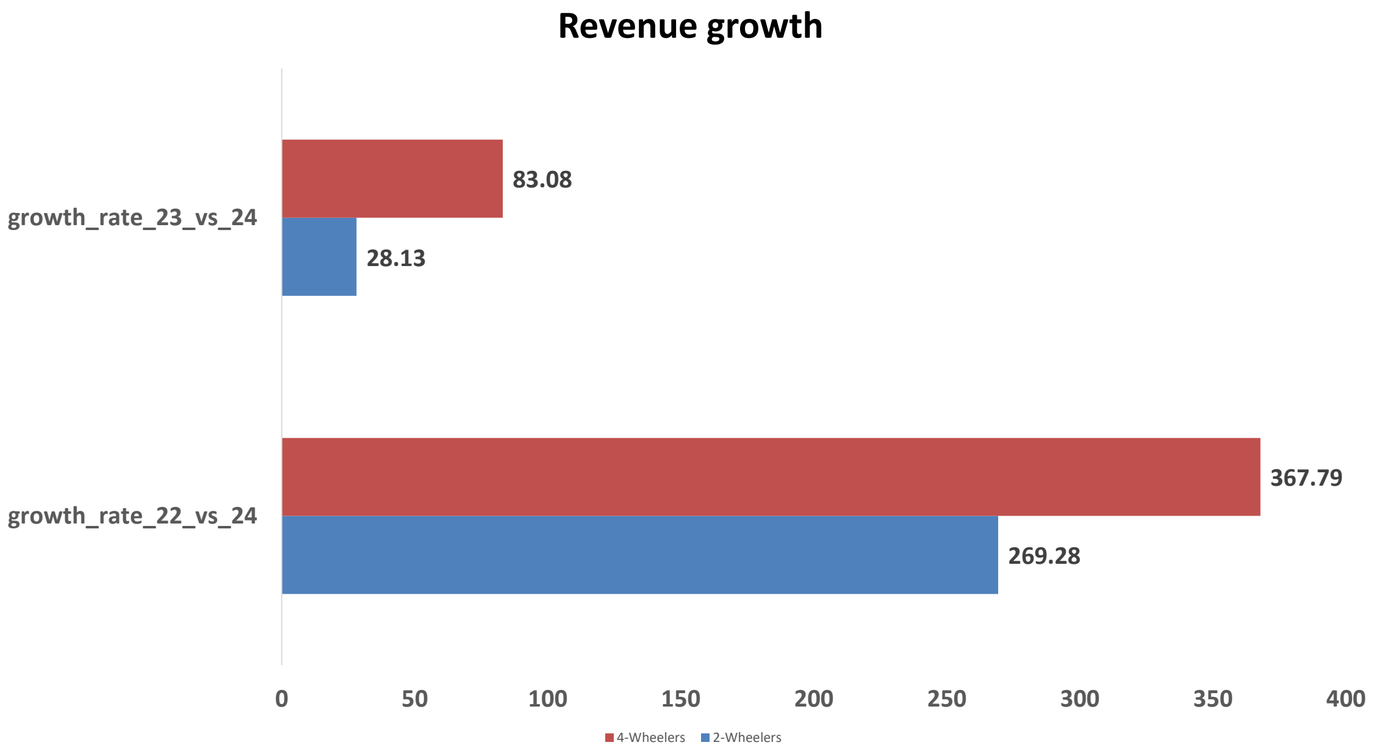
Q9: EV sales projection for 2030 (Top 10 states by penetration)

- Peak Sales Performance (March): March is consistently the highest-performing month for EV sales across all ana years, significantly outperforming other months with a three-year aggregate of approximately 291,587 units.
- Low Sales Performance (June and May): The lowest sales typically occur in the first quarter of the fiscal year, with June and May recording the lowest aggregate volumes (approximately 106,709 and 159,869 units respectively).
- Fiscal Year Consistency: The "Peak" in March remained stable from 2022 through 2024, while the "Low" period shifted slightly from May in fiscal years 2022 and 2023 to June in fiscal year 2024.



Q10: Revenue growth analysis (2W & 4W EVs)

•**2-Wheelers drive volume growth**, with sales rising significantly and dominating overall EV adoption
• **4-Wheelers lead in revenue and faster growth**, showing strong expansion in the premium segment
• **Market split is clear:** 2W = volume-driven, while 4W = value-driven with higher growth momentum



Secondary Research Questions

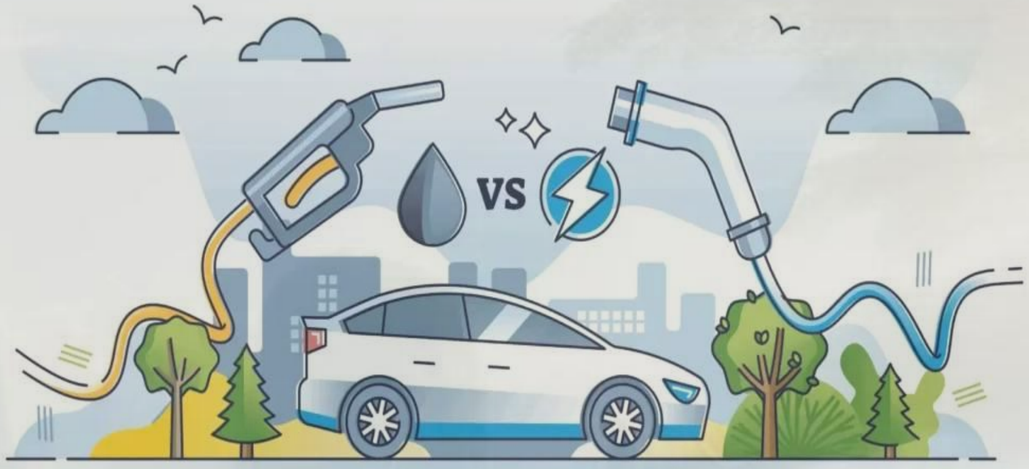
1. What are the primary reasons for customers choosing 4-wheeler EVs in 2023 and 2024 (cost savings, environmental concerns, government incentives)?
2. How do government incentives and subsidies impact the adoption rates of 2-wheelers and 4-wheelers? Which states in India provided most subsidies?
3. How does the availability of charging stations infrastructure correlate with the EV sales and penetration rates in the top 5 states?
4. Who should be the brand ambassador if AtliQ Motors launches their EV/Hybrid vehicles in India and why?
5. Which state of India is ideal to start the manufacturing unit? (Based on subsidies provided, ease of doing business, stability in governance etc.)
6. top 3 recommendations for AtliQ Motors.

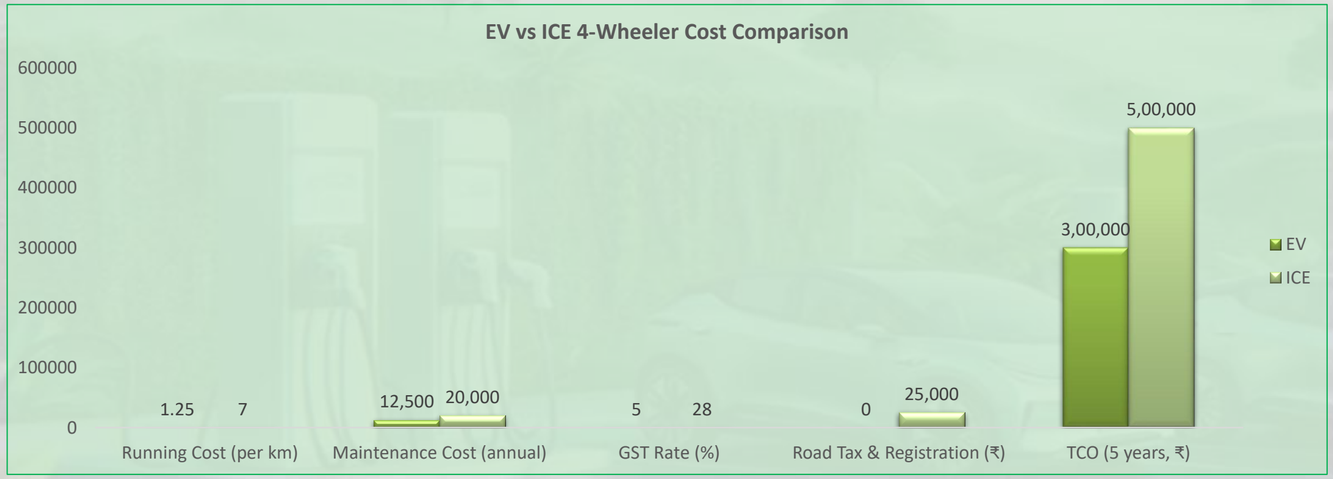
Qn 1: the primary reasons for customers choosing 4-wheeler EVs in 2023 and 2024 (cost savings, environmental concerns, government incentives)

Factor	Description	Impact on Customers
Lower Running Cost	Electricity is cheaper than petrol/diesel; fewer moving parts reduce servicing costs	Major cost savings in long-term ownership
Environmental Awareness	EVs produce zero tailpipe emissions and reduce carbon footprint	Appeals to eco-conscious buyers
Government Incentives	Subsidies under FAME II Scheme + tax benefits	Reduces initial purchase cost
Rising Fuel Prices	Petrol/diesel prices continue to increase	EVs become more economical over time
Tech Advancements	Better battery range, smart features, fast charging	Improves driving experience & trust
Urban Suitability	Ideal for daily commuting & city travel	Convenient for home charging users
Social Influence	Early adopters & status symbol effect	Attracts young professionals
Model Availability	More EV options across price segments	Increased consumer choice

Lower Running Cost

Category	EV Cost	ICE Cost(Internal Combustion Engine)	Notes
Running Cost (per km)	**₹1 – ₹1.5**	₹6 – ₹8	**EVs are 4–6x cheaper per km**
Maintenance Cost (annual)	₹10,000 – ₹15,000	₹15,000 – ₹25,000	**EVs have fewer moving parts, reducing service costs**
Government Incentives	GST 5%, subsidies available, tax benefits	GST 28% no subsidies	**EVs benefit from central/state incentives**
Road Tax & Registration	Exempt or minimal	Applicable (varies by state)	**Many states waive EV road tax**
Total Cost of Ownership (5 years)	Lower overall despite higher upfront cost	Higher due to fuel & maintenance	**EVs save money long-term**





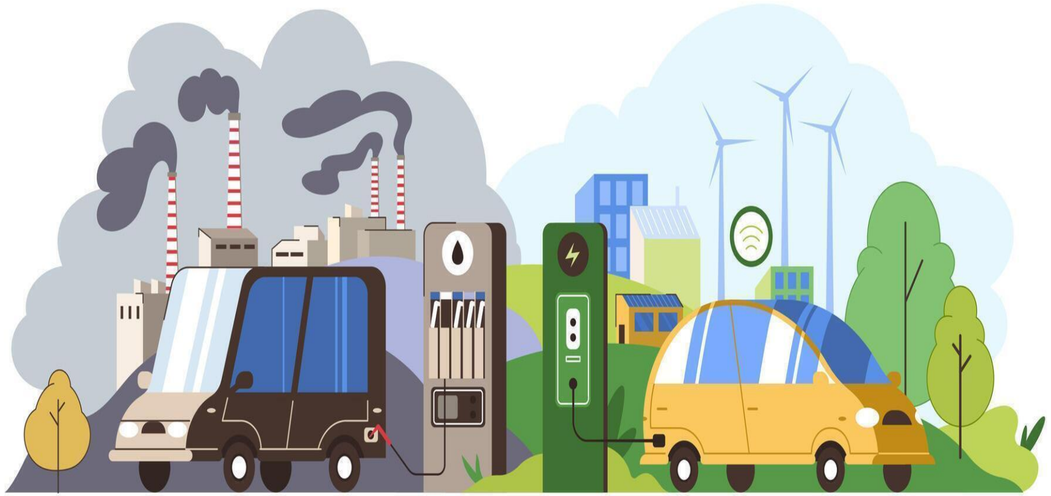
- **Fuel savings dominate**: EVs cost about **₹1/km vs ₹6–8/km** for ICE cars.
- **Maintenance savings**: EVs are **30–40% cheaper** annually.
- **Policy support**: EVs enjoy **lower GST (5%)**, subsidies, and road tax exemptions.
- **Overall ownership**: Over 5 years, EVs typically have a **lower total cost of ownership** despite higher upfront prices.

Source

Environmental Awareness

Aspect	Electric Vehicles (EVs)	Petrol/Diesel Vehicles (ICE)
Tailpipe Emissions	**Zero** – no CO_2, NOx, or particulate matter	High CO_2, NOx, PM emissions → urban air pollution
Carbon Footprint (lifecycle)	30–50% lower (depends on electricity mix)	High lifecycle emissions from fuel extraction, refining, combustion
Noise Pollution	Very low – quiet operation	Significant engine noise and vibration
Climate Impact	Helps reduce greenhouse gases, supports climate goals	Major contributor to global warming
Air Quality	Improves urban air quality, reduces smog	Worsens air quality, linked to respiratory issues
Consumer Appeal	Strong among eco-conscious buyers	Increasingly seen as unsustainable

Category	EV (Electric Vehicle)	Petrol/Diesel Vehicle
Production Emissions	9–10 tCO_2e	6–7 tCO_2e
Annual Driving Emissions	1.8 tCO_2e	4.5 tCO_2e
Lifetime Emissions (10 yrs)	27–30 tCO_2e	50–55 tCO_2e
Tailpipe Emissions		-0~4.5 tCO_2e per year



• EVs **produce zero tailpipe emissions**, making them far cleaner in cities.
• Even accounting for battery production, EVs typically cut **lifetime CO_2 emissions by ~40%** compared to ICE vehicles.
• Environmental awareness is now a **top 3 driver of EV adoption in India**, with ~39% of surveyed buyers citing it as a key factor.

Source

Government Incentives

Incentive Type	Example	Impact on Buyers
FAME II Subsidy	Direct purchase subsidy (~20%)	Lowers upfront cost
Tax Benefits	₹1.5 lakh deduction on loan interest	Reduces financing burden
State Incentives	Road tax exemption, fee waiver	Improves affordability regionally

Factor	Electric Vehicles (EVs)	Petrol/Diesel Vehicles (ICE)
National Subsidy	FAME II scheme: up to **20% cost reduction**	**None**
Tax Benefits	Section 80EEB: **₹1.5 lakh deduction** on EV loan interest	**None**
State Incentives	Road tax exemption, registration fee waiver, extra subsidies	**None**
Upfront Cost Impact	Reduced significantly due to combined incentives	No reduction; fuel taxes increase cost
Policy Support	Strong central + state backing for adoption	No direct support, higher regulatory burden
Market Effect	EV penetration rising (**6.5% of total sales in 2023**)	Petrol/diesel share declining gradually

FAME II Subsidy (20% cost reduction)
•Covers electric two-wheelers, three-wheelers, and four-wheelers.
•Focuses on public transport (e-buses) and shared mobility.
•Directly lowers upfront purchase cost, making EVs more competitive with petrol/diesel.

Tax Benefit (80EEB, ~10% reduction)
•Allows individuals to claim up to ₹1.5 lakh deduction on interest paid for EV loans.
•Encourages middle-class buyers to finance EVs affordably.
•Reduces long-term ownership cost.

State Incentives (~8% reduction)
•Road tax exemptions and registration fee waivers in states like Delhi, Maharashtra, Tamil Nadu.
•Some states also offer additional subsidies for two-wheelers and charging infrastructure.

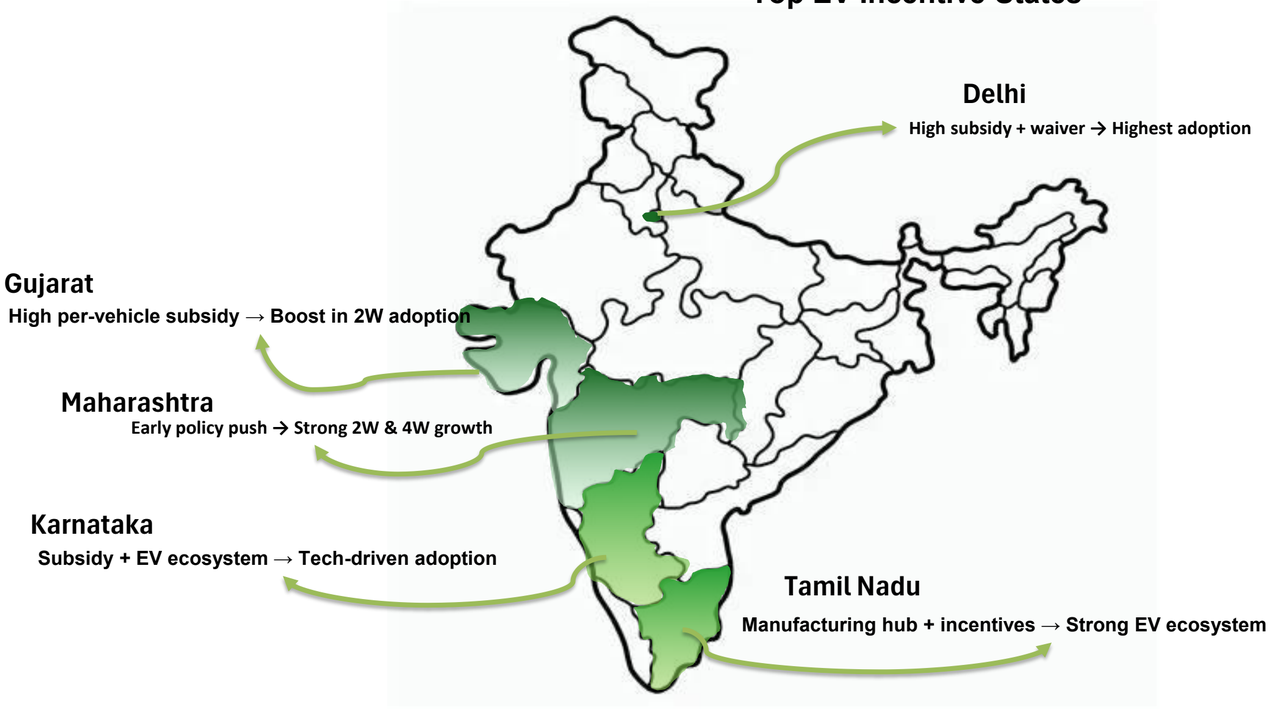
Source

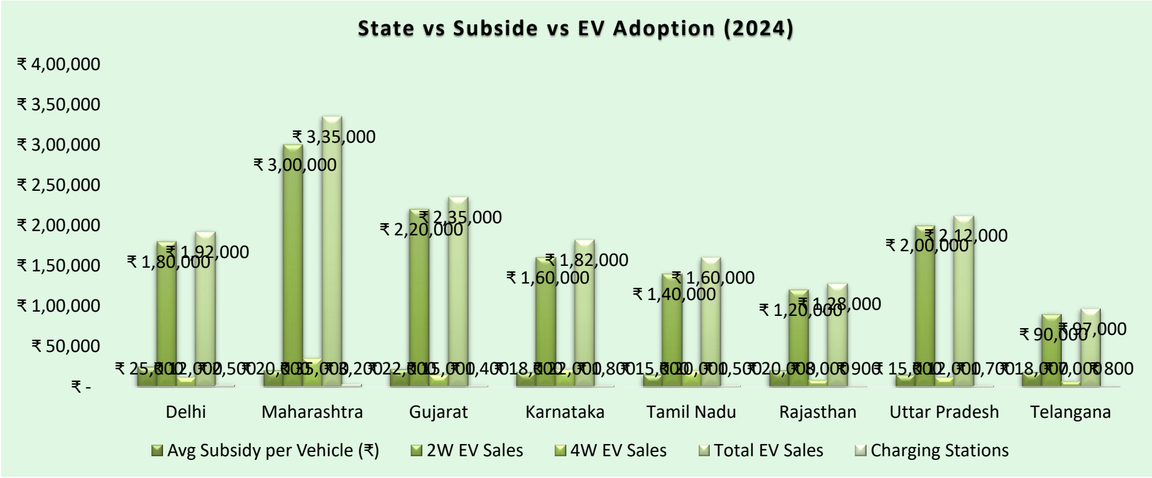
Qn 2 : How do government incentives and subsidies impact the adoption rates of 2-wheelers and 4-wheelers? Which states in India provided most subsidies?

Impact of Government Incentives on EV Adoption (2W vs 4W)

Aspect	2-Wheeler EVs (2W)	4-Wheeler EVs (4W)
Subsidy Impact	High impact – reduces price gap significantly	Moderate impact – still expensive even after subsidy
Sales Growth	Strong growth due to subsidies	Gradual growth due to high upfront cost
Adoption Sensitivity	Very sensitive to subsidy changes	Less sensitive (income segment higher)
Key Scheme	FAME II Scheme boosted demand significantly	FAME-II + state incentives support adoption
Data Insight	+12.7% sales increase per subsidy boost	211% higher growth in states with incentives
State Policy Effect	54.5% higher sales in states with incentives	Strong growth where state subsidies exist

Top EV Incentive States





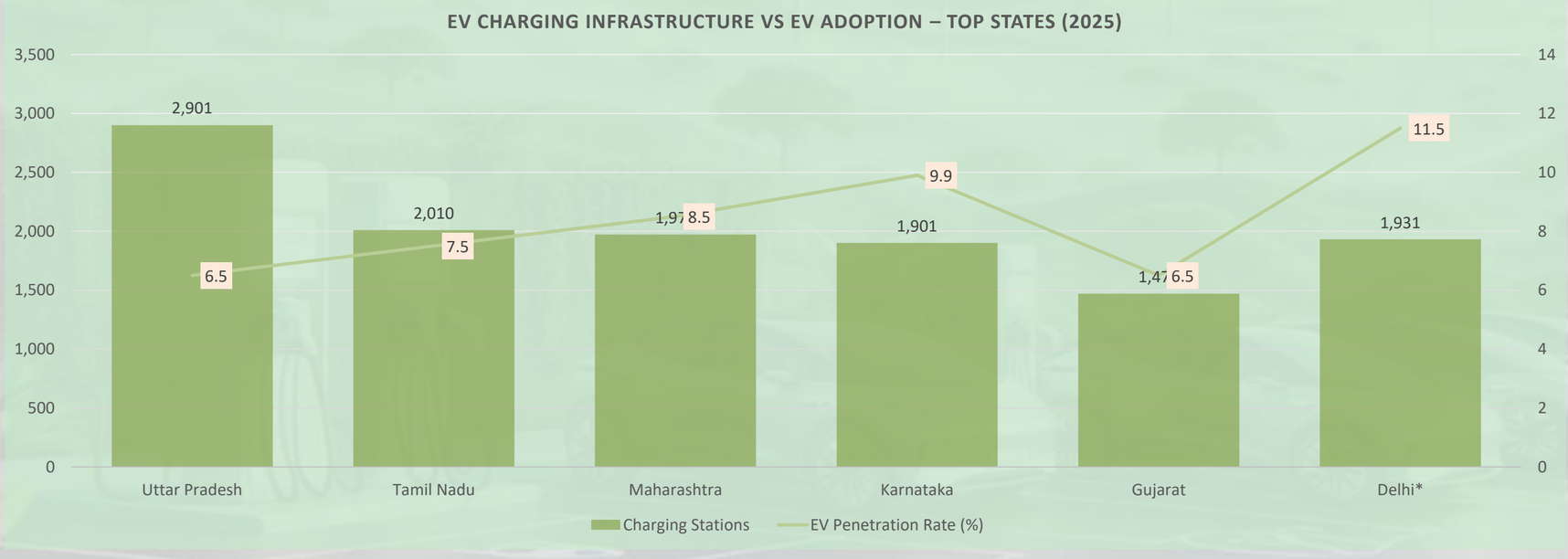
- Subsidies directly reduce upfront cost → biggest barrier removed.
- 2W EV adoption is **highly dependent** on subsidies.
- 4W EV adoption depends on **cost + technology + infrastructure**.
- States with incentives recorded **up to 54.5% higher EV adoption**, especially in 2-wheelers.

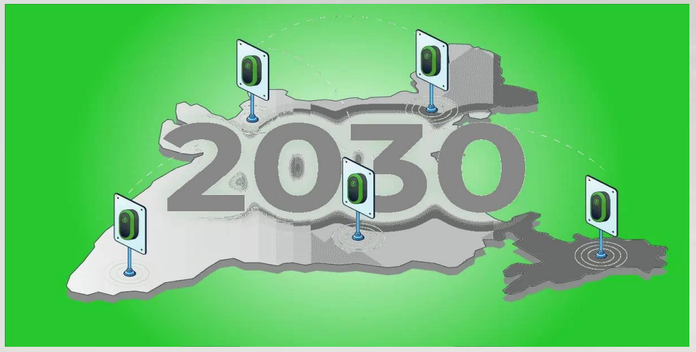
Source

Qn 3: How does the availability of charging stations infrastructure correlate with the EV sales and penetration rates in the top 5 states?

Correlation Between Charging Infrastructure & EV Adoption

Rank	State	Charging Stations	EV Penetration Rate	Key Observations
1	**Uttar Pradesh**	**2,901**	~6–7%	Infrastructure leader, but penetration moderate → infra alone isn't enough.
2	**Tamil Nadu**	**2,010**	~7.5%	Manufacturing hub; infra + ecosystem drive steady adoption.
3	**Maharashtra**	**1,972**	~8.5%	Strong EV base; chargers heavily utilized, especially for 4W.
4	**Karnataka**	**1,901**	~9.9%	Dense network (esp. Bengaluru); infra strongly correlates with penetration.
5	**Gujarat**	**1,470–1,500**	~6.5%	Subsidy-driven 2W adoption, supported by infra.
—	**Delhi** *(outside top 5 infra)*	~1,931	**11.50%**	Penetration leader due to aggressive subsidies + tax waivers, despite fewer chargers.





Infrastructure vs Adoption

•**UP**: Infrastructure leader, but penetration moderate (~6–7%).
•**Delhi**: Penetration leader (11.5%) despite not being in top 5 for chargers.
•**Tamil Nadu**: Second in chargers (2,010), steady adoption (~7.5%) thanks to manufacturing ecosystem.
•**Maharashtra**: Strong EV base (1,972 chargers, ~8.5% penetration), chargers heavily utilized.
•**Karnataka**: Dense network (1,901 chargers, ~9.9% penetration), strong correlation between infra and adoption.
•**Gujarat**: ~1,470 chargers, ~6.5% penetration, subsidy-driven adoption.

Correlation Takeaway:

•Infrastructure alone doesn't guarantee adoption (UP).
•Subsidies can offset infra gaps (Delhi).
•Balanced states (MH, KA, TN, GUJ) show how infra + ecosystem sustain growth.

Source

Qn 4: Who should be the brand ambassador if AtliQ Motors launches their EV/Hybrid vehicles in India and why?

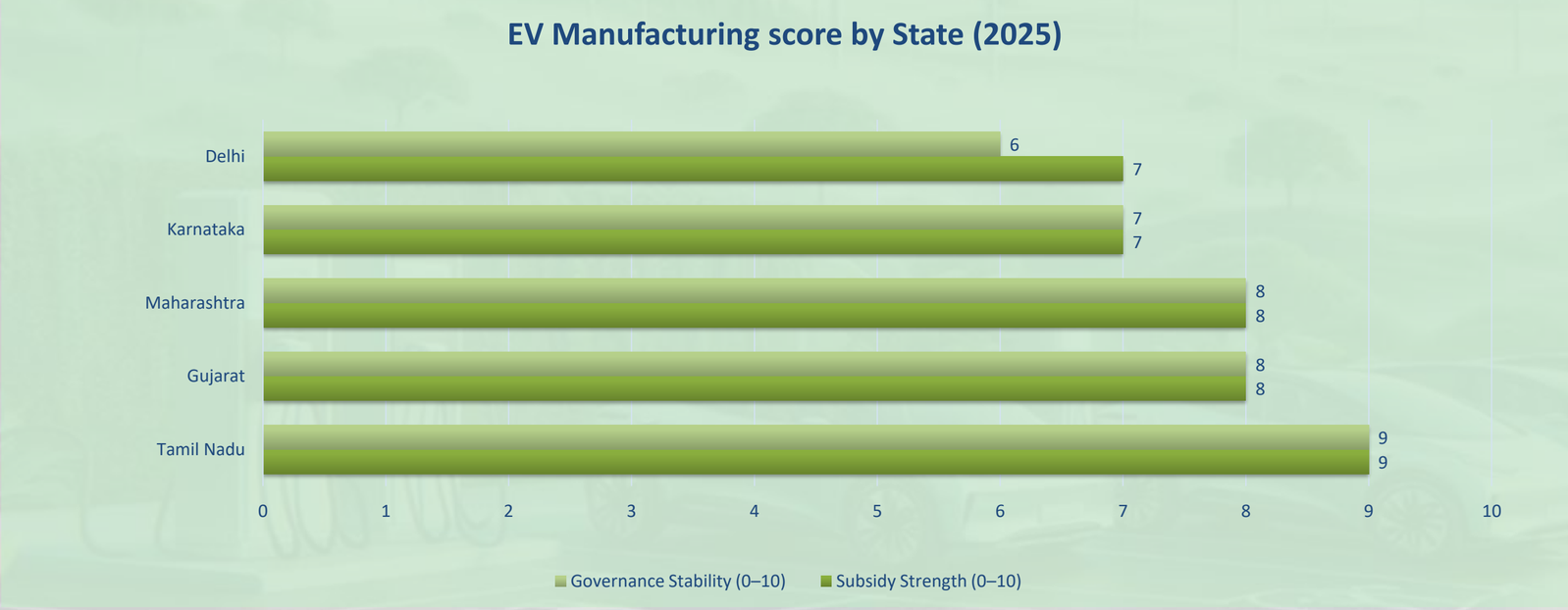
brand ambassador for AtliQ Motors

•**Vicky Kaushal** is the ideal brand ambassador for AtliQ Motors due to his strong association with EVs, relatable personality, and appeal among urban consumers, making him a perfect fit to promote sustainable and innovative mobility solutions in India.

Qn 5 : Which state of India is ideal to start the manufacturing unit? (Based on subsidies provided, ease of doing business, stability in governance etc.)

Strategic Location Analysis for EV Manufacturing in India

State	Business Environment Score (2025)	Rank	Key Strengths
Tamil Nadu	**7.4**	**1**	Industrial hub (automobiles, electronics, renewables), strong policies, stable governance
Gujarat	7.3	2	Investor magnet, robust industrial base, strong infrastructure
Maharashtra	7.1	3	Economic powerhouse, auto hub, ambitious MSME & infra investments
Karnataka	6.9	4	IT & biotech hub, sunrise sectors (AI, green hydrogen), balanced growth
Delhi	6.9	4	Startup magnet, finance & IT, infrastructure modernization



Tamil Nadu leads India's 2025 Business Environment Rankings with a score of 7.4, offering the strongest combination of subsidies, ease of doing business, and governance stability , making it the most strategic choice for AtliQ Motors' EV manufacturing unit.

Source

Qn 6 Top 3 recommendations for AtliQ Motors.

Top 3 Recommendations for AtliQ Motors

Strategic actions to accelerate growth and strengthen position in India's EV market

1

Establish Manufacturing in Tamil Nadu

- Strong automotive ecosystem and supplier base
- Skilled workforce and technical talent availability
- Excellent port connectivity for exports
- Attractive incentives and investor-friendly policies

IMPACT

Lower production cost
+ faster scalability

2

Focus on Affordable 2W & Entry-Level EV Segment

- 2-wheelers drive majority of EV adoption in India
- Price-sensitive market with high growth potential
- Government subsidies boosting demand
- Quick market penetration and higher sales volume

IMPACT

Higher market share
+ quicker growth

3

Invest in Charging Infrastructure Partnerships

- Charging infrastructure is key to EV adoption
- Reduces range anxiety and builds customer trust
- Partner with leading players like
  - **Tata Power**
  - **Ather Energy**

IMPACT

Stronger ecosystem
+ long-term brand loyallty